

HDOC - Day 31

2D Array Operations

Algorithm

Read the matrix once. Option 1 sums every element using nested loops. Option 2 calculates each row sum and stores it in a 1D array. Option 3 displays the transpose by interchanging row/column traversal. Repeat the menu until Exit.

Test Case Table

T C	Input/Operation	Expected	Actual	Result
1	2x3: 1..6, op1	21	21	Pass
2	same, op2	[6,15]	[6,15]	Pass
3	same, op3	1 4 / 2 5 / 3 6	Matches	Pass

C Program

```
#include <stdio.h>
int main(){int r,c,i,j,ch,sum;printf("Enter rows and columns: ");scanf("%d%d",&r,&c);int a[r][c],rs[r];
printf("Enter elements: ");for(i=0;i<r;i++)for(j=0;j<c;j++)scanf("%d",&a[i][j]);
do{printf("\n1.Sum all\n2.Row sums\n3.Transpose\n4.Exit\nEnter choice: ");scanf("%d",&ch);
switch(ch){case 1:sum=0;for(i=0;i<r;i++)for(j=0;j<c;j++)sum+=a[i][j];printf("Sum of all elements = %d\n",sum);break;
case 2:for(i=0;i<r;i++){rs[i]=0;for(j=0;j<c;j++)rs[i]+=a[i][j];printf("Sum of row %d = %d\n",i+1,rs[i]);}
printf("Row-sum array:  [");for(i=0;i<r;i++)printf("%d%s",rs[i],i==r-1?"":", ");printf("]\n");break;
case 3:printf("Transpose:\n");for(j=0;j<c;j++){for(i=0;i<r;i++)printf("%d ",a[i][j]);printf("\n");}break;
case 4:break;default:printf("Invalid choice.\n");}while(ch!=4);return 0;}
```

Compilation, Execution & Testing

```
puppie@Anvikas-MacBook-Air Desktop % clang day31.c -o day31
puppie@Anvikas-MacBook-Air Desktop % ./day31
Rows columns: 2 3 Elements: 1 2 3 4 5 6
Choice: 1
Sum of all elements = 21
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day31
Choice: 2
Sum of row 1 = 6
Sum of row 2 = 15
Row-sum array: [6, 15]
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day31
Choice: 3
Transpose:
1 4
2 5
3 6
```

Result: The program was compiled, executed and tested successfully.